coefficient of viscosity
সান্দ্রতা সহগ / সান্দ্রতা গুণাঙ্ক

viscosity = সান্দ্রতা → stickiness → s^{-1}

বেগ অবক্রম / বেগ গ্রেডিয়েন্ট:
একটি তরলের বেগ কত দ্রুত একটি নির্দিষ্ট দিকে পরিবর্তিত হচ্ছে তার একটি পরিমাপ।

Principles of Physics I (PHY111)

Lab

Experiment no: 5

Name of the experiment: Determination of the coefficient of viscosity of a given liquid by using Stoke's law

YOU HAVE TO BRING A GRAPH PAPER (cm scale) TO DO THIS EXPERIMENT.

Theory

In this experiment we determine the coefficient of viscosity of a liquid (glycerin) by using Stoke's law.

Buoyancy
প্লবতা

We gently place a ball on the surface of the liquid. As the ball is immersed into the liquid three different forces act on the ball. These are:

i) Weight of the ball, F_w which acts vertically downward

ii) Buoyancy force of the liquid on the ball, F_b which acts vertically upward and

iii) Drag force on the ball due to viscosity of the liquid, F_d which always acts against the direction of motion, in this case vertically upward since the ball moves downward.

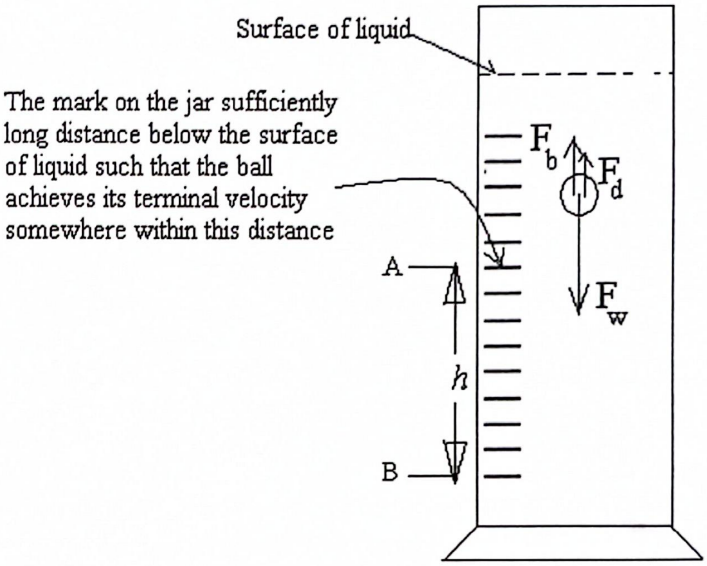


Figure 1: Arrangement for the experiment

Let the radius of the ball = r, then the volume of the ball is $V = \frac{4}{3}\pi r^3$ and the mass of the ball is $m = \rho V$, where ρ is the density of the ball's material.

So, the weight of the ball is $F_w = mg = \rho Vg = \rho\left(\frac{4}{3}\pi r^3\right) g$ (1)

সান্দ্রতা: → PI (poiseiulle) → Nsm^{-2}
তরল পদার্থের অভ্যন্তরীন ঘর্ষণের ফলে প্রবাহে (বয়ে যেতে) বাধা দেবার প্রবণতাকে সান্দ্রতা বলা হয়।

সান্দ্রতা গুণাঙ্ক: নির্দিষ্ট তাপমাত্রায় প্রবাহীর দুটি পাশাপাশি স্তরের মধ্যে একক বেগ-অবক্রম বজায় রাখতে প্রবাহী স্তরের প্রতি একক ক্ষেত্রফলে যে স্পর্শকীয় বলের প্রয়োজন হয় তাকে উক্ত তাপমাত্রায় প্রবাহীর সান্দ্রতা গুণাঙ্ক বা সান্দ্রতা সহগ বলে।

Let, the density of liquid is σ. Using Archimedes principle buoyancy force of liquid on the ball is given by, $F_b = \sigma \left(\frac{4}{3} \pi r^3 \right) g$ (2)

If v is the velocity of the ball and η is the coefficient of viscosity of the liquid, then according to Stoke's law the drag force due to viscosity, $F_d = 6 \pi r \eta v$ (3)

When the ball attains terminal velocity, the net force is zero. So upward force must be equal to the downward force, i.e.,

$$F_w = F_b + F_d \quad (4)$$

Using (1), (2), (3) and (4) we can show that

$$\eta = \frac{2}{9} \frac{r^2}{v_t} (\rho - \sigma) g \quad (5)$$

In this experiment we used different balls of different radii, r but made of same material (ρ is same) and for each of them we measure v_t. Then we draw a graph by plotting r^2 along x –axis and corresponding v_t along y-axis. It should be a straight line passing through the origin. The slopes of the line gives us the value of $\frac{v_t}{r^2}$. So from (5)

$$\eta = \frac{2}{9} \frac{(\rho - \sigma) g}{slope} \quad (6)$$

After drawing the v_t vs. r^2 graph and working out its slope we can deduce the coefficient of viscosity of glycerin by using equation (6)

Apparatus

A jar of glycerin, balls of different radii, screw gauge, stop watch and a meter scale.

Procedure

1. Write down the density of glycerin, σ and density of the ball's material, ρ in section A and B of the data sheet respectively.

2. Fix a mark, A (as shown in figure 1) on the jar sufficiently long distance below the surface of glycerin such that the ball achieves its terminal velocity while passing this mark (say, it is the mark specifying 600 ml). Fix another mark, B (as shown in figure 1) near the bottom of the jar (say, it is the mark specifying 100 ml). Measure the distance, h between two marks and write it down in the table 1 of section C of data sheet.

3. Take a ball and measure its diameter by using a screw-gauge. Please see to know how the use a screw-gauge appendix A (given in the softcopy of this script which is available in the server). Find out its radius and write it down in the table-1 of section C.

4. Take the stop watch in one hand and with other hand gently place the ball at the middle of the top surface of the liquid. Make sure that the ball does not touch the wall while it is moving downward.

5. When the ball passes the mark A, then turn on the stop watch. When the ball passes the mark B, then turn off the stop-watch. Time recorded in the stop watch is, t. Write it down in table-1 of section C.

6. Find out the terminal velocity, $v_t = h/t$

7. Repeat steps 2, 3 and 4 for other balls.

8. Draw v_t vs. r^2 graph and work out the slope of the graph.

9. Determine the value of η.

Read carefully and follow the following instructions:

- **Please READ the theory carefully, TAKE printout of the 'Questions on Theory' and ANSWER the questions in the specified space BEFORE you go to the lab class.**
- **To get full marks for the 'Questions on Theory' portion, you must answer ALL of these questions CORRECTLY and with PROPER UNDERSTANDING, BEFORE you go to the lab class. However, to ATTEND the lab class you are REQUIRED to answer AT LEAST the questions with asterisk mark.**
- **Write down your NAME, ID, THEORY SECTION, GROUP, DATE, EXPERIMENT NO AND NAME OF THE EXPERIMENT on the top of the first paper.**
- **If you face difficulties to understand the theory, please meet us BEFORE the lab class. However, you must read the theory first.**
- **DO NOT PLAGIARIZE. Plagiarism will bring ZERO marks in this WHOLE EXPERIMENT. Be sure that you have understood the questions and the answers what you have written, and all of these are your own works. You WILL BE asked questions on these tasks in the class. If you plagiarize for more than once, WHOLE lab marks will be ZERO.**
- **After entering the class, please submit this portion before you start the experiment.**

Name: Mohammad Nahiyan Rahman **ID:** 24101157 **Sec:** 22 **Group:** 3 **Date:** 25.2.24

Experiment no: 5

Name of the Experiment: Determination of the coefficient of viscosity of a given liquid by using Stoke's law

Questions on Theory

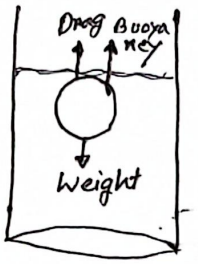
*1) When the ball falls downward in the liquid then what are the major forces which act on the ball? Draw the free body diagram. [0.5]

Ans: When the ball falls downward in the liquid, three ~~mg~~ major forces act on the ball.

i) Weight of the ball.

ii) Buoyancy force of the liquid.

iii) Drag force due to ~~viscos~~ viscosity of the liquid.



*2) State the law of Archimedes to find out the buoyancy force acting on the ball. [0.5]

Ans: Archimede's principle states that the upward buoyant force that is exerted on an object submerged in a fluid, whether fully or partially, is equal to the weight of the fluid that the object displaces.

$F_W = F_B + F_d$

$\Rightarrow F_B = F_W - F_d = W_{fl}$

F_B = Buoyancy Force

F_d = Drag force of the liquid

F_W = Weight of the ball

W_{fl} = Weight of fluid displaced

*3) Find an expression of the weight of the ball in terms of the density of ball's material and its radius. [0.5]

Ans:

$\cancel{F_B = \sigma\left(\frac{4}{3}\pi r^3\right)g}$

$\cancel{\Rightarrow F_W = \sigma\left(\frac{4}{3}\pi r^3\right)g}$

F_W = Weight of the ball

$F_W = mg = \rho V g = \rho\left(\frac{4}{3}\pi r^3\right)g$

*4) Find an expression of the buoyancy force on the ball by the liquid in terms of the density of liquid and the radius of the ball. [0.5]

Ans: $F_b = \sigma\left(\frac{4}{3}\pi r^3\right)g$ σ = density of glycerin

*5) If v is the velocity of the ball, η is the coefficient of viscosity of the liquid and r is the radius of the ball, then according to Stoke's law what is the drag force on the ball due to viscosity? [0.5]

Ans: Drag force, $F_d = 6\pi r\eta v$

*6) What is the terminal velocity of the ball? [0.25]

Ans: Terminal velocity, $V_t = \frac{F_d}{6\pi r\eta}$

*7) When the terminal velocity is achieved by the ball then write down the equation relating the major forces acting on the ball. [0.25]

Ans: $F_W = F_b + F_d$

8) Starting from this equation show that, $\eta = \frac{2}{9}\frac{r^2}{v_t}(\rho - \sigma)g$ [Use additional page(s) if you require]

[2]

Ans: $F_W = F_b + F_d$

$\Rightarrow \frac{4}{3}\pi r^3 \rho g = \frac{4}{3}\pi r^3 \sigma g + 6\pi r\eta V_t$

$\Rightarrow 6\pi r\eta V_t = \frac{4}{3}\pi r^3 \rho g - \frac{4}{3}\pi r^3 \sigma g$

$\Rightarrow 6\pi r\eta V_t = \frac{4}{3}\pi r^3 g(\rho - \sigma)$

$\Rightarrow \eta V_t = \frac{4\pi}{3\times 6\pi} \times \frac{r^3}{r} g(\rho - \sigma)$

$\Rightarrow \eta = \frac{2r^2}{9V_t}(\rho - \sigma)g$